



# LINEAR ALGEBRA AND ITS APPLICATIONS

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## CLASS-6

### PROPERTIES OF EIGEN VALUES AND EIGEN VECTORS AND CAYLEY-HAMILTON THEOREM

## PROPERTIES OF EIGEN VALUES AND EIGEN VECTORS:

- If  $\lambda$  is an Eigen value of  $A$  with  $x$  as the corresponding Eigen vector then  $\lambda^2$  is an Eigen value of  $A^2$  with the same Eigen vector  $x$ .
- For a given Eigen vector  $x$ , there corresponds only one Eigen value  $\lambda$ .
- For a given Eigen value there corresponds infinitely many Eigen vectors.

- $\lambda = 0$  is an Eigen value of  $A$ , if and only if  $A$  is singular i.e  $\det(A)=0$ .
- If  $\lambda$  is an Eigen value of  $A$  with  $x$  as the Eigen vector then  $1/\lambda$  is an Eigen value of  $A^{-1}$  provided  $A^{-1}$  exists.
- $A$  and its transpose  $A^T$  have the same Eigen values.

- The Eigen values of a diagonal matrix are just the diagonal elements of the matrix.
- The Eigen values of an idempotent matrix are either zero or unity.
- The sum of the Eigen values of a matrix is the sum of the elements of the principal diagonal.
- The product of the Eigen values of a matrix  $A$  is equal to its determinant.

“Every square matrix satisfies its own characteristic equation”

Proof: Let  $A$  be an  $n$ -square matrix. Let  $D(\lambda)$  be the characteristic polynomial of  $A$  given by,

$$D(\lambda) = |\lambda I - A| = \lambda^n + C_{n-1}\lambda^{n-1} + C_{n-2}\lambda^{n-2} + \dots + C_1\lambda + C_0 \longrightarrow \textcircled{1}$$

Let  $B(\lambda)$  be the adjoint of  $(\lambda I - A)$ . The elements of  $B(\lambda)$  are cofactors of the matrix  $(\lambda I - A)$  and are polynomials in  $\lambda$  of degree not exceeding  $n-1$ . Thus

$$B(\lambda) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \dots + B_1\lambda + B_0 \longrightarrow \textcircled{2}$$

where  $B_i$  are  $n$ -square matrices whose elements are functions of the elements of  $A$  and independent of  $\lambda$ .

We know that  $(\lambda I - A) \cdot \text{adj}(\lambda I - A) = (\lambda I - A)I$

$$(\lambda I - A) \cdot B(\lambda) = (\lambda I - A)I$$

From (1) & (2), we have

$$\begin{aligned} (\lambda I - A)(B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \dots + B_1\lambda + B_0) \\ = I(\lambda^n + C_{n-1}\lambda^{n-1} + \dots + C_1\lambda + C_0) \longrightarrow (3) \end{aligned}$$

Equating the like powers of  $\lambda$  on both sides of (3), we get

$$B_{n-1} = I$$

$$B_{n-2} - AB_{n-1} = C_{n-1}I$$

$$B_{n-3} - AB_{n-2} = C_{n-2}I$$

.....

$$B_0 - AB_1 = C_1I$$

$$-AB_0 = C_0I$$

Multiplying both sides of the above matrix equations by  $A^n, A^{n-1}, A^{n-2}, \dots, A, I$ , respectively, we have



$$A^n B_{n-1} = A^n$$

$$A^{n-1} B_{n-2} - A^n B_{n-1} = c_{n-1} A^{n-1}$$

$$A^{n-2} B_{n-1} - A^{n-1} B_{n-2} = c_{n-2} A^{n-2}$$

.....

$$A B_0 - A^2 B_1 = c_1 A$$

$$- A B_0 = c_0 I$$

By adding all the above equations, we get

$$0 = A^n + c_{n-1} A^{n-1} + c_{n-2} A^{n-2} + \dots + c_1 A + c_0 I \longrightarrow \textcircled{4}$$

Since all the terms on the L.H.S cancel each other, thus  $A$  satisfies its own characteristic equation.

Inverse by Cayley-Hamilton theorem:-

Multiplying (4) by  $A^{-1}$ ,

$$0 = A^{n-1} + c_{n-1}A^{n-2} + c_{n-2}A^{n-3} + \dots + c_1I + c_0A^{-1}$$

Solving for  $A^{-1}$ , we get

$$A^{-1} = -\frac{1}{c_0} [A^{n-1} + c_{n-1}A^{n-2} + \dots + c_1I]$$

Note:-  $A^{-1}$  exists only if  $c_0 = \text{determinant of } A \neq 0$ .



**THANK YOU**

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